

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I VAISHNAVI.S (192521313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

A retail chain has implemented a Smart Retail Inventory Management System using cloud computing to efficiently manage inventory across multiple stores. Traditionally, inventory was maintained separately in each store, leading to problems such as stock shortages, overstocking, delayed updates, and inaccurate inventory records. By moving to a cloud-based solution, the retail chain can monitor and control inventory in real time from a centralized location.

The system consists of Point of Sale (POS) systems, web dashboards, cloud services, cloud storage, and analytics platforms. Whenever a customer purchases a product, the POS system sends the transaction details to the cloud through secure Application Programming Interfaces (APIs). The cloud instantly updates the inventory database, ensuring that stock information is synchronized across all stores. Managers can access dashboards from any location to monitor inventory levels, sales performance, and product availability.

The cloud storage service securely stores product details, supplier information, customer transaction logs, sales history, and analytics datasets. Since the data is stored in the cloud, it can be accessed quickly and backed up automatically, reducing the risk of data loss. The system also uses Platform as a Service (PaaS) for data analytics and demand forecasting. Advanced analytics tools examine historical sales data, seasonal trends, and customer purchasing patterns to predict future demand. This enables the retailer to restock products at the right time, reduce waste, and improve customer satisfaction. Security is an important part of the system. A Virtual Private Network (VPN) provides encrypted communication between store branches and cloud servers. Firewalls protect the network from unauthorized access and cyber threats, while an API Gateway authenticates and manages API requests, ensuring that only authorized users and devices can access cloud services. These security measures protect sensitive business and customer information.

The cloud infrastructure provides high scalability, allowing the retail chain to add new stores or increase computing resources without purchasing additional hardware. It also offers high availability, automatic backups, and disaster recovery, ensuring uninterrupted business operations. Employees can access the system through web applications or mobile devices, making inventory management faster and more efficient. Overall, the Smart Retail Inventory Management System demonstrates how Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) work together in a cloud environment. The solution improves inventory accuracy, enhances decision-making through analytics, reduces operational costs, strengthens security, and increases customer satisfaction. By integrating cloud infrastructure, services, and web applications, the retail chain achieves a modern, scalable, secure, and efficient inventory management system that supports business growth and operational excellence.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

A large enterprise has implemented a Cloud-Based Document Collaboration System that allows employees to create, edit, share, and manage documents online, similar to Google Docs. Earlier, employees worked on separate copies of documents, resulting in duplicate files, version conflicts, and communication delays. By adopting a cloud-based collaboration platform, multiple users can work on the same document simultaneously from different locations, improving teamwork and productivity. The system is accessed through web browsers without requiring complex software installation. Users simply log in with their credentials and can create new documents or edit existing ones from laptops, desktops, tablets, or smartphones. The platform supports real-time collaboration, where changes made by one user are instantly visible to others. This feature allows teams to work together efficiently regardless of their physical location.

All documents are stored in distributed cloud storage, which ensures that data is replicated across multiple servers and data centers. This provides high availability, fault tolerance, and protection against hardware failures. The system also includes a version control mechanism, which automatically saves every change made to a document. Users can view previous versions, restore older copies if required, and track who made specific changes. This prevents accidental data loss and makes document management more reliable.

The collaboration platform uses Web APIs to synchronize document updates across multiple devices and users. Whenever a user edits a document, the changes are transmitted through APIs to the cloud server, which immediately updates all connected clients. This synchronization process ensures that every user always views the latest version of the document, even when accessing it from different devices.

Security is a major component of the collaboration system. Encryption protects documents during transmission and while stored in the cloud, preventing unauthorized access. Identity and Access Management (IAM) verifies user identities through secure authentication methods such as usernames, passwords, and multi-factor authentication. Access permissions allow document owners to control whether other users can view, comment on, or edit specific documents. These security features protect confidential business information while enabling safe collaboration. Overall, the Cloud-Based Document Collaboration System demonstrates how cloud computing supports real-time collaboration, distributed storage, API-based synchronization, global accessibility, scalability, and strong security. By combining Infrastructure as a Service (IaaS) for cloud resources, Platform as a Service (PaaS) for application development and synchronization, and Software as a Service (SaaS) for end-user document editing, the enterprise achieves a secure, reliable, and highly efficient collaboration environment that enhances productivity, simplifies document management, and supports modern digital workplaces.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time

An emerging startup has developed a Cloud-Based Online Food Ordering Platform that allows customers to browse restaurant menus, place orders, make payments, and track deliveries using web browsers and mobile applications. The platform connects customers, restaurants, delivery partners, and administrators through a centralized cloud environment. Traditional food ordering methods often involve manual processes, delayed communication, and limited scalability. By adopting cloud computing technologies, the startup can provide fast, reliable, and efficient services to users across multiple locations. The system architecture consists of frontend clients, backend services, cloud storage, virtual machines, container orchestration platforms, and security mechanisms. Customers access the platform using mobile applications or web browsers. These frontend applications provide a user-friendly interface where users can search restaurants, view menus, add items to the cart, make online payments, and monitor delivery status. Restaurant owners use separate dashboards to update menus, manage orders, and monitor sales. Delivery partners also use mobile applications to receive delivery requests and update order status in real time.

Communication between frontend applications and backend services is achieved using RESTful Web APIs. APIs act as a bridge between clients and cloud services by exchanging data in formats such as JSON. When a customer places an order, the application sends a request to the backend API, which processes the order, validates payment details, updates restaurant records, and notifies delivery personnel. APIs also handle user authentication, menu retrieval, payment processing, and order tracking. This API-driven architecture ensures smooth communication and enables integration with external payment gateways, mapping services, and notification systems.

The platform uses cloud storage services to securely store menu images, invoices, customer details, transaction records, and restaurant information. Menu images and documents are stored in object storage systems, while customer information and order records are stored in cloud databases. Cloud storage provides scalability, automatic backup, and high availability, ensuring that data remains accessible even during hardware failures. Data replication across multiple servers enhances reliability and supports disaster recovery. The startup deploys its applications using Virtual Machines (VMs) and Managed Kubernetes services. Virtual machines provide the required computing resources, operating systems, and networking environments for application hosting. However, as the number of users increases, managing applications manually becomes difficult. Therefore, the startup uses Kubernetes, a container orchestration platform, to automate deployment, scaling, and management of application containers. The platform follows a microservices architecture, where different services such as user management, restaurant management, payment processing, order handling, and delivery tracking operate independently. Kubernetes automatically distributes workloads across multiple servers, performs load balancing, and

scales services according to user demand. During peak hours, such as weekends or holidays, additional containers are created automatically to handle increased traffic. If a server fails, Kubernetes shifts workloads to healthy nodes, ensuring high availability and uninterrupted service.

Security is a critical aspect of the food ordering platform because it handles customer data, payment information, and business records. The system uses authentication tokens, such as JSON Web Tokens (JWT), to verify user identity and maintain secure sessions. Once users log in, a token is generated and attached to subsequent requests, ensuring that only authenticated users can access services.

The platform uses HTTPS (Hypertext Transfer Protocol Secure) to encrypt communication between clients and cloud servers. Encryption prevents attackers from intercepting sensitive information such as passwords, payment details, and personal data. The system also implements Role-Based Access Control (RBAC), where different users are assigned specific permissions. Customers can place and track orders, restaurant managers can update menus and view sales reports, delivery agents can access delivery information, and administrators can manage the overall platform. RBAC ensures secure and controlled access to resources. To provide real-time functionality, the platform integrates notifications, live order tracking, and instant updates. Customers receive notifications about order confirmation, preparation status, dispatch, and delivery. Restaurant owners receive order alerts immediately, while delivery agents get navigation information through integrated map services. These real-time features improve user experience and operational efficiency. The cloud platform also offers monitoring and analytics services that help administrators track system performance, server utilization, customer preferences, and sales trends. Analytics data enables the startup to make better business decisions, optimize delivery operations, and improve customer satisfaction. Automated monitoring tools generate alerts when performance issues occur, ensuring proactive maintenance. The Online Food Ordering Platform demonstrates how modern cloud technologies integrate clients, APIs, storage systems, virtual machines, Kubernetes, and security services to build scalable and reliable applications. It utilizes Infrastructure as a Service (IaaS) for virtual machines and networking resources, Platform as a Service (PaaS) for Kubernetes and application management, and Software as a Service (SaaS) for delivering food ordering services to customers.

In conclusion, the cloud-based food ordering system provides scalability, flexibility, security, and high availability. The use of RESTful APIs enables seamless communication, cloud storage ensures reliable data management, Kubernetes provides automatic scaling, and security mechanisms protect sensitive information. By integrating these technologies, the startup can deliver a fast, efficient, and secure food ordering experience while supporting business growth and customer satisfaction.